

QUESTION 1 (15 Marks)**Marks**

- (a) By “completing the square” or otherwise, find the minimum value of $2x^2 - 4x + 5$ for all real values of x . 2
- (b) A radioactive substance of mass M decays at a rate proportional to its mass present, ie; $M = M_0 e^{kt}$.
Its initial mass is 400 grams and reduces to 300 grams after 2 years.
- (i) How many grams have decayed at the end of four years? 4
- (ii) The half-life of the substance is the time taken for the mass to be reduced by half. What is the half-life of the substance, (to the nearest month). 3
- (c) The equation of motion of an object moving x cm about a fixed point O , after t seconds, along a straight line is given by $x = \tan t$ for $0 \leq t < \frac{\pi}{2}$.
- (i) Express its velocity $v(t)$ and acceleration $a(t)$ in terms of t . 3
- (ii) Show that its acceleration is $a(t) = 2x(1 + x^2)$. 3

QUESTION 2 (15 Marks)

- (a) For what values of k is $kx^2 + (3+k)x + (3+k)$ positive definite. 4
- (b) Nine marbles numbered 1, 2, 3, 4, 5, 6, 7, 8 & 9 are placed in a bag and three are drawn out at random (each not replaced). What is the probability that the sum of the numbers on the three marbles drawn is odd. 3
- (c) The equation of motion of a particle moving x metres along a straight line after t seconds is given by $x(t) = 2t^3 - 6t^2 + 3$.
- (i) What is the particle’s initial speed and acceleration. 2
- (ii) When and where does it first come to rest. 2
- (iii) Sketch a velocity/time graph and briefly explain what is happening to the motion when $t = 1$ second. 2
- (iv) Briefly describe the motion of the particle 2

QUESTION 3 (15 Marks)**Marks**

- (a) The population of two colonies after t years is given by $P_1 = 2000e^{0.138t}$ and $P_2 = 5000e^{0.04t}$. The initial population of each was recorded on the 1st January, 2006.
- (i) How long will it take for the population P_1 to triple.
(Answer to the nearest year.) 2
- (ii) Calculate the year and month when both populations are the same. 3
- (iii) Calculate the rate at which P_1 is increasing at this time. 2
- (b) Two chess players, Bostik and Spastik, play three games of chess to contest a win. In any game they play, the probability that Bostik wins is $\frac{5}{10}$ and the probability that Spastik wins is $\frac{4}{10}$.
What is the probability that Bostik wins the competition. 3
- (c) An object moves x cm along a straight line after t seconds with its velocity function $v = -e^{-2t}$ for $t \geq 0$ and is initially at the origin.
- (i) Derive an expression for its displacement as a function of time. 3
- (ii) Neatly sketch $x = f(t)$ for $t \geq 0$. 2

QUESTION 4 (15 Marks)

- (a) Find all real values of t for which the quadratic equation $\frac{x^2 - x + 1}{x^2 + x + 1} = t$ has real and different roots. 4
- (b) A particle of unit mass is projected vertically upwards from a point O with a velocity of 25 m/s and has an acceleration of -10 m/s².
- (i) Find its velocity and height above O after a time $t > 0$. 3
- (ii) Find its maximum height of projection. 2
- (b) A deck of cards contains the red Jacks, Queens, Kings and Aces from a normal pack of 52 playing cards (ie; the Hearts and Diamonds). Jenny is dealt two cards from this deck of cards. What is the probability that:
- (i) She has two Aces if she announces that she has at least one Ace. 2
- (ii) She has a pair if she announces that she does not have the Ace of Hearts. 2
- (iii) She has two cards of the same suit if she announces that she has at least one Heart and one King. 2

END of PAPER

YEAR 12 TERM 2
H.S.C. ASSESSMENT 2006

SOLUTIONS

1/ (a) $2x^2 - 4x + 5$
 $= 2(x^2 - 2x + \frac{5}{2})$
 $= 2(x^2 - 2x + 1 + \frac{3}{2})$
 $= 2(x-1)^2 + 3$

$\therefore$ MIN of 3 at $x=1$

OR using $x = -\frac{b}{2a}$
 $x = 1 \therefore 2(1)^2 - 4(1) + 5$
 $= 3$

(b)(i) $M = M_0 e^{kt}$
 $= 400 e^{kt}$

When $t=2$, $M=300$

$\therefore 300 = 400 e^{2k} \therefore k = \frac{1}{2} \ln \frac{3}{4}$

When $t=4$ $\frac{3}{4} \ln \frac{3}{4}$

$M = 400 e^{\frac{3}{2} \ln \frac{3}{4}}$
 $= 400 \left(\frac{3}{4}\right)^2$
 $= 225$

$\therefore$ 175 gms. decayed.

(c)(i) $x = \tan t$

$\therefore v(t) = \sec^2 t$ — (1)

$\therefore a(t) = 2 \sec t \cdot \sec t \tan t$

$\therefore a(t) = 2 \sec^2 t \tan t$ — (2)

(ii) When $M = \frac{1}{2} M_0$ $\frac{1}{2} \ln \frac{3}{4}$

$\therefore \frac{1}{2} M_0 = M_0 e^{\frac{t}{2} \ln \frac{3}{4}}$
 $\therefore \ln \frac{1}{2} = \frac{t}{2} \ln \frac{3}{4}$

$\therefore t = \frac{2 \ln 0.5}{\ln 0.75}$

(ii) $a(t) = 2 \tan t (1 + \tan^2 t)$

$\therefore t = 4.82$ years

$\therefore a(t) = 2x(1+x^2)$

$= 4$ years 10 months.

since $x = \tan t$

2/ (a) $Kx^2 + (3+K)x + (3+K)$

We require $K > 0$ and $\Delta < 0$

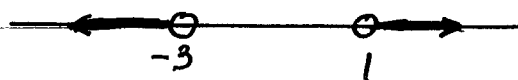
$\Delta = (3+K)^2 - 4K(3+K)$

$= (3+K)(3+K-4K)$

$= (3+K)(3-3K)$

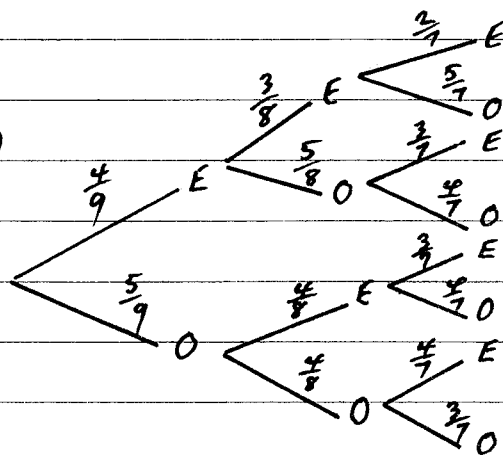
$= 3(3+K)(1-K)$

For $\Delta < 0$



For $K > 0$ we require
only $K > 1$.

(b)



$P = P(E+E+O) + P(O+O+O)$

$= 4 \left(\frac{4}{9} \times \frac{3}{8} \times \frac{5}{7} \right)$

$= \frac{4 \times 5}{3 \times 2 \times 7}$

$\therefore P = \frac{10}{21}$

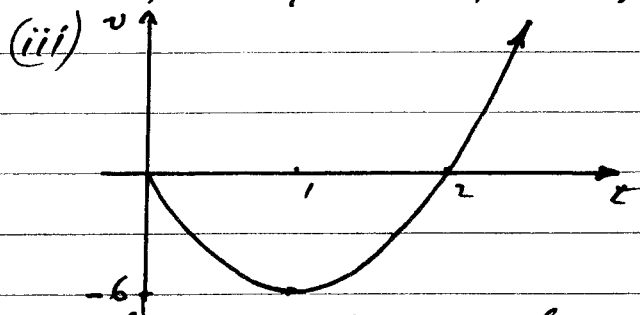
2 (c)(i) $x(t) = 2t^3 - 6t^2 + 3$

$v(t) = 6t^2 - 12t$

$a(t) = 12t - 12$

When $t = 0$

$v(0) = 0 \text{ m/s}$ and $a(0) = -12 \text{ m/s}^2$



When $t = 1$ it reaches maximum speed OR zero force acting on the particle

(ii) It comes to rest when $v = 0$ ie; when $t = 0$ or $t = 2$. Since $t > 0$ $\therefore$ when $t = 2$ and $x = -5$ ie; 5 metres left of 0.

(iv) The particle starts 3m right of 0 increasing its speed for 1 sec. then slows down and stops 5m left of 0 and passing through 0 between 2 and 3 seconds. It then increases its speed moving right of 0 indefinitely.

3 (a)(i) $P_1 = 2000 e^{0.138t}$

$P_2 = 5000 e^{0.04t}$

When $P_1 = 4000$

(ii) When $P_1 = P_2$

(iii) $\frac{dP_1}{dt} = 2000 \times 0.138 e^{0.138t}$

$4000 = 2000 e^{0.138t}$

$2000 e^{0.138t} = 5000 e^{0.04t}$

When $t = 9.349$

$\therefore \ln 2 = 0.138t$

$\therefore e^{0.098t} = \frac{5}{2}$

$\frac{dP_1}{dt} = 276 e^{1.290162}$

$\therefore t = 5.023$

$\therefore 0.098t = \ln 2.5$

$\frac{dP_1}{dt} = 1002.8$

$\therefore t = 5 \text{ years (nearest year)}$

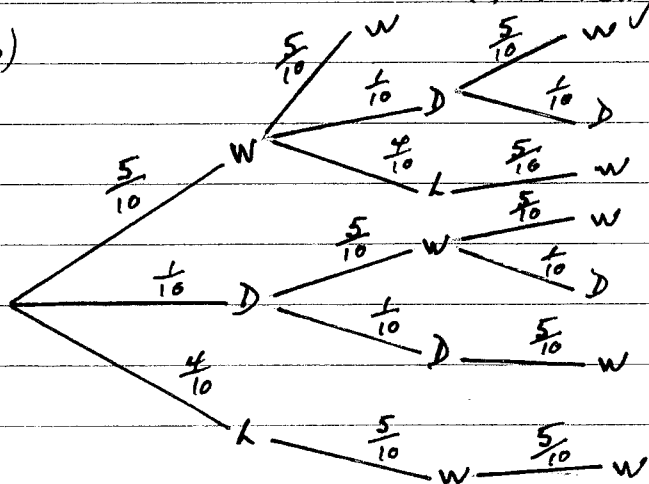
$\therefore t = 9.349$

$= 9 \text{ years } 5 \text{ months}$

$\approx 1003 / \text{year}$

$\therefore$ During May 2015

(b)



$$P = \left(\frac{5}{10} \times \frac{5}{10}\right) + \left(\frac{5}{10} \times \frac{1}{10} \times \frac{5}{10}\right) + \left(\frac{5}{10} \times \frac{1}{10} \times \frac{1}{10}\right) + \left(\frac{5}{10} \times \frac{4}{10} \times \frac{5}{10}\right) + \left(\frac{1}{10} \times \frac{5}{10} \times \frac{5}{10}\right) + \left(\frac{1}{10} \times \frac{5}{10} \times \frac{1}{10}\right) + \left(\frac{4}{10} \times \frac{1}{10} \times \frac{5}{10}\right) + \left(\frac{4}{10} \times \frac{5}{10} \times \frac{5}{10}\right)$$

$$= \frac{25}{100} + \frac{25}{1000} + \frac{5}{1000} + \frac{100}{1000} + \frac{135}{1000}$$

$\therefore P = \frac{103}{200}$

3. (c) (i) $v = \frac{dx}{dt} = -e^{-2t}$

$$\therefore x = -\int e^{-2t} dt$$

$$\therefore x = \frac{1}{2} e^{-2t} + C$$

When $t=0$, $x=0$

$$\therefore 0 = \frac{1}{2} + C \therefore C = -\frac{1}{2}$$

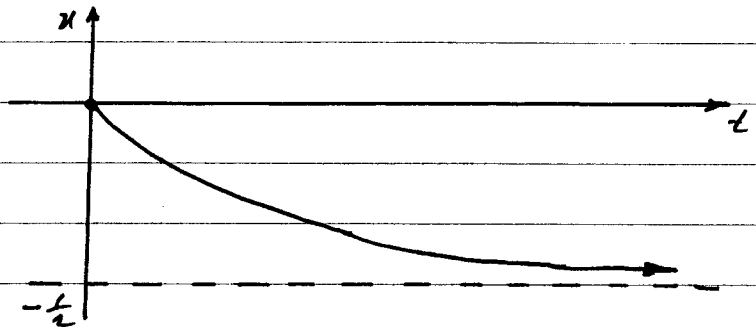
$$\therefore x = \frac{1}{2} e^{-2t} - \frac{1}{2}$$

$$\therefore x = \frac{1}{2} (e^{-2t} - 1)$$

(ii) When $t=0$, $x=0$

$$\text{Let } x = \frac{1}{2} \left(\frac{1}{e^{2t}} - 1 \right)$$

As $t \rightarrow \infty$, $\frac{1}{e^{2t}} \rightarrow 0 \therefore x \rightarrow -\frac{1}{2}$



4. (a) $\frac{x^2 - x + 1}{x^2 + x + 1} = t$

$$x^2 + x + 1$$

$$\therefore x^2 - x + 1 = t(x^2 + x + 1)$$

$$\therefore (t-1)x^2 + (t+1)x + (t-1) = 0$$

For real and different roots, we require $\Delta > 0$

$$\text{Now } \Delta = (t+1)^2 - 4(t-1)^2$$

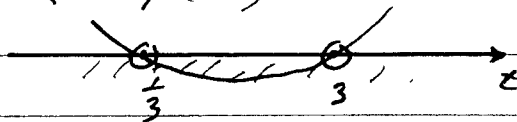
$$= t^2 + 2t + 1 - 4t^2 + 8t - 4$$

$$= -3t^2 + 10t - 3$$

$$= -3(t^2 - 10t + 3)$$

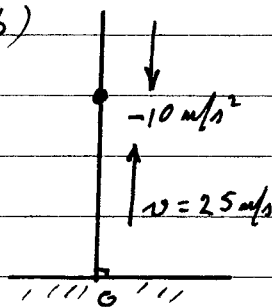
For $\Delta > 0$

$$(3t-1)(t-3) < 0$$



$$\therefore \frac{1}{3} < t < 3$$

(b)



$$v = \frac{dx}{dt} = 25$$

$$\therefore x = \int 25 dt$$

$$\therefore x = 25t + C$$

When $t=0$, $x=0 \therefore C=0$

$$\therefore x = 25t$$

Now $\frac{dv}{dt} = -10$

$$\therefore v = -\int 10 dt$$

$$\therefore v = -10t + C$$

When $t=0$, $v=0 \therefore C=0$

$$\therefore v = -10t$$

(i) Now $v = 25 - 10t$ (since $v \uparrow = 25 \text{ m/s}$)

$$\therefore \frac{dx}{dt} = 25 - 10t$$

$$\therefore x = \int (25 - 10t) dt$$

$$\therefore x = 25t - 5t^2 + C$$

When $t=0$, $x=0 \therefore C=0$

$$\therefore x = 25t - 5t^2$$

(ii) Maximum height when $v=0$, i.e., when $t = \frac{5}{2}$

When $t = \frac{5}{2}$, $x = 25\left(\frac{5}{2}\right) - 5\left(\frac{5}{2}\right)^2$

$$\therefore x = 31\frac{1}{4} \text{ m.}$$

4 (c) (i)

A_D	*	*	*	*	*	*	*	⊗
A_H	*	*	*	*	*	*	*	⊗
K_D							*	*
K_H							*	*
Q_D							*	*
Q_H							*	*
J_D							*	*
J_H							*	*
	J_H	J_D	Q_H	Q_D	K_H	K_D	A_H	A_D

$$\rho = \frac{2}{26} = \frac{1}{13}$$

(ii)

A_D	*	*	*	*	*	*	*	*
A_H	*	*	*	*	*	*	*	*
K_D	*	*	*	*	⊗	*	*	*
K_H	*	*	*	*	*	⊗	*	*
Q_D	*	*	⊗	*	*	*	*	*
Q_H	*	*	*	⊗	*	*	*	*
J_D	⊗	*	*	*	*	*	*	*
J_H	*	⊗	*	*	*	*	*	*
	J_H	J_D	Q_H	Q_D	K_H	K_D	A_H	A_D

$$\rho = \frac{6}{42} = \frac{1}{7}$$

(iii)

A_D					*			
A_H					⊗	*		
K_D	*		*		*		*	
K_H	⊗	*	⊗	*	*		⊗	*
Q_D					*			
Q_H					⊗	*		
J_D					*			
J_H					⊗	*		
	J_H	J_D	Q_H	Q_D	K_H	K_D	A_H	A_D

$$\rho = \frac{6}{20} = \frac{3}{10}$$